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Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Personal history with meta-analysis (Meyer et al., 2002, 2012)
- Challenges to conducting comprehensive reviews
- AI to the Rescue
 - Where we are
 - Where we are going
 - What is possible
- Envisioning a Living Meta-analytic Database

Literature search method

Two complementary search strategies in every database

- **Keyword & subject-term search** — truncated organizational terms (org*, work*, employ*, job*, firm*, institution*, affect*, norm*, continu*) within three words of commit*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search** — every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only** — academic journals, books, dissertations & theses, conference papers, reports, and working papers.
- **One exception** — ERIC does not support cited-reference searching, so it received the keyword search only (13 searches in total).

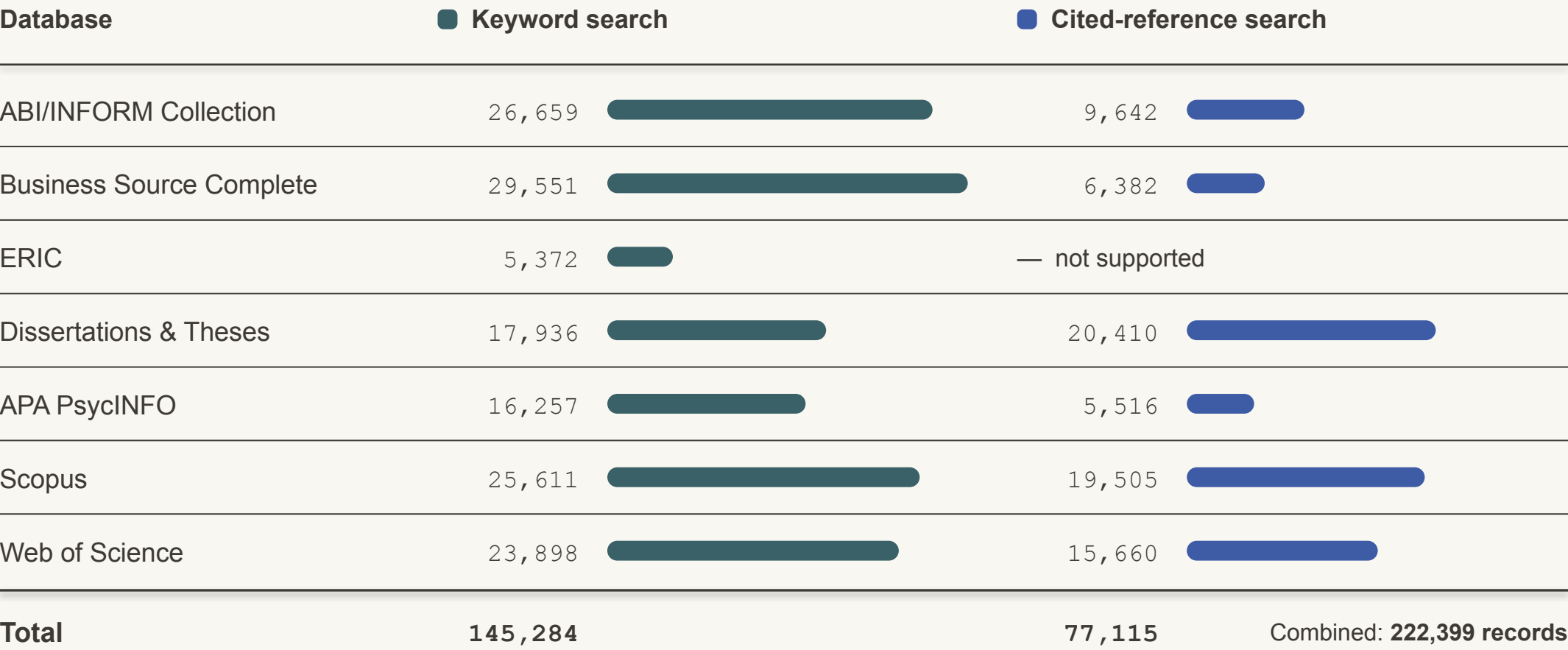
Seven databases

Searched December 20, 2025 – January 7, 2026

ABI/INFORM Collection	ProQuest
Business Source Complete	EBSCO
ERIC	Ovid · keyword only
Dissertations & Theses	ProQuest
APA PsycINFO	Ovid
Scopus	Elsevier
Web of Science	Clarivate

Records identified, by database and search strategy

Records retrieved per database, before consolidation and de-duplication · bars share one scale



ERIC (Ovid) does not support cited-reference searching.

From 222,399 records to 22,484 unique abstracts

All records retrieved — 13 searches across 7 databases

222,399



▼ consolidated and de-duplicated



22,484

unique abstracts

10.1% of the initial yield — the abstracts behind every analysis that follows

How accurate is AI screening? — from prior research

Before any coding, every record has to be judged relevant or not — the first filter, deciding which studies even enter the review. In earlier published validation, the AI was compared against the expert human teams who screened by hand.

■ Screening — deciding which studies belong

97.2%

of relevant studies — vs **91.6%** for humans

5 meta-analyses · 30,150 abstracts screened

■ What the AI is being compared to

The human baseline was thorough — every record judged by hand:

2×

two reviewers + a 3rd

→

1 pass

the AI

≈ **98–99% less time** — days or months of work, done in minutes to an hour.

- **Data extraction** — pulling every number out of each study — was just as strong: the AI matched expert coders at **97.0%** (humans 92.1%) across 20,476 data points.

WHAT THE CODED ABSTRACTS REVEAL

Abstract Analysis

How the literature has grown — and what it studies.



How quickly has the literature on organizational commitment grown? Here's the annual count of articles in our collection.

Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)



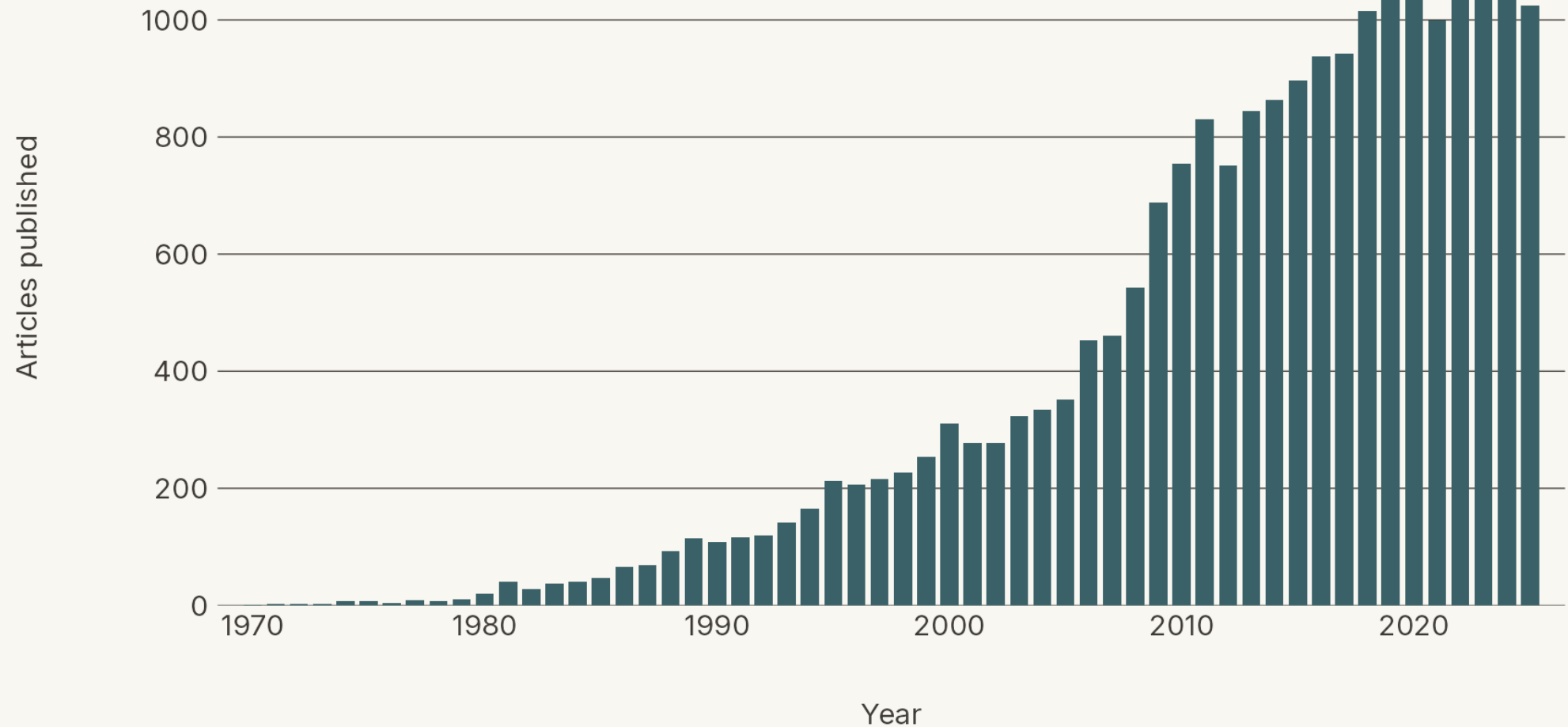
Articles Published Per Year

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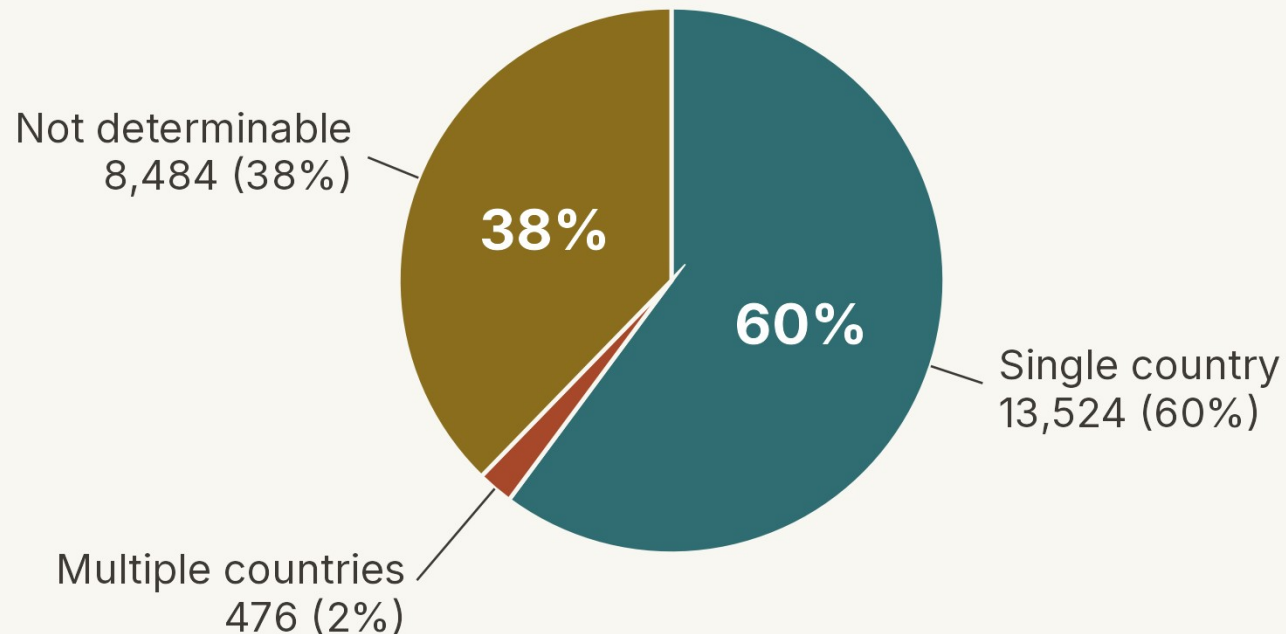
Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)



How many of the articles can we place geographically?

Country-coding status across all 22,484 unique abstracts



How country of origin was determined

Country coding criteria (each YES / NO):

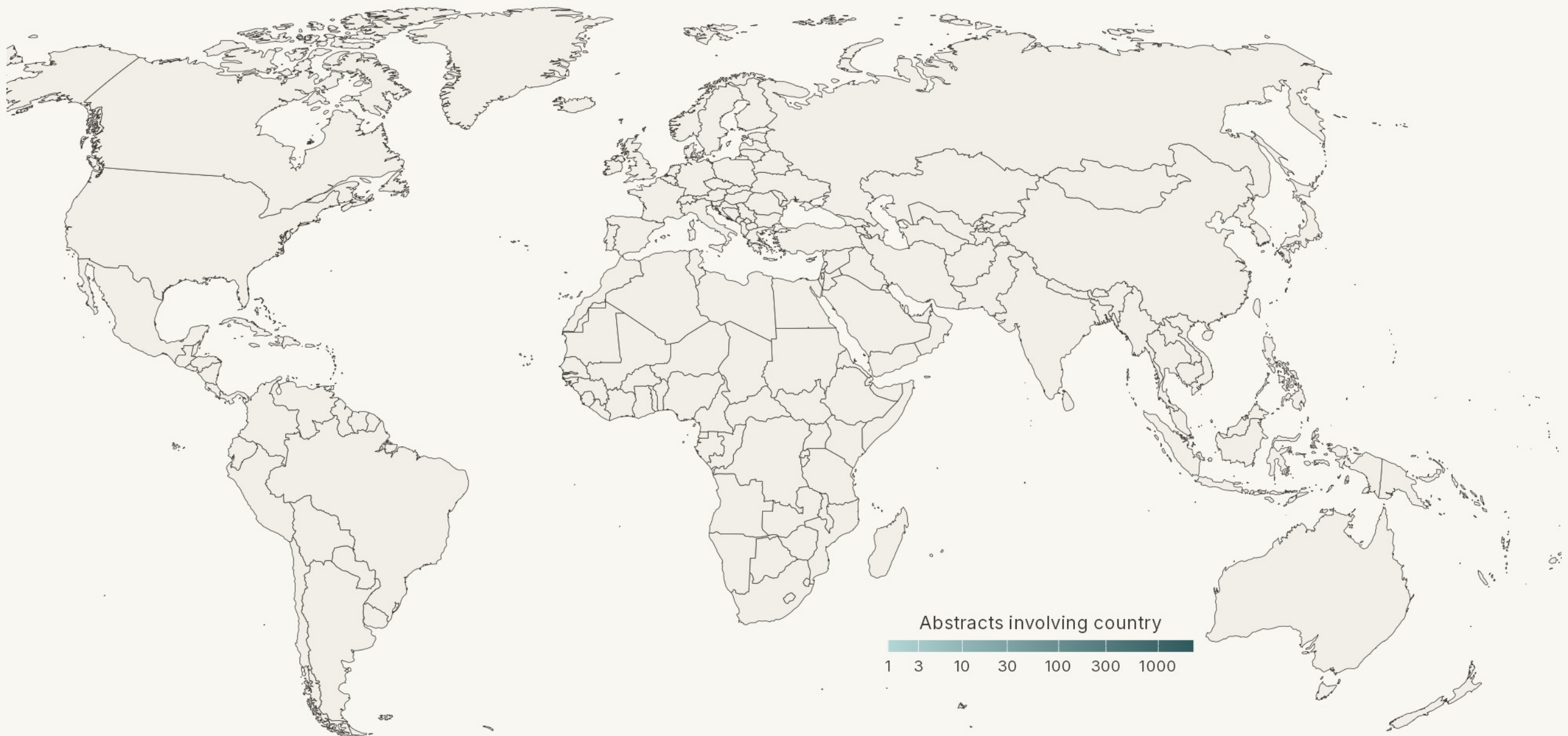
- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

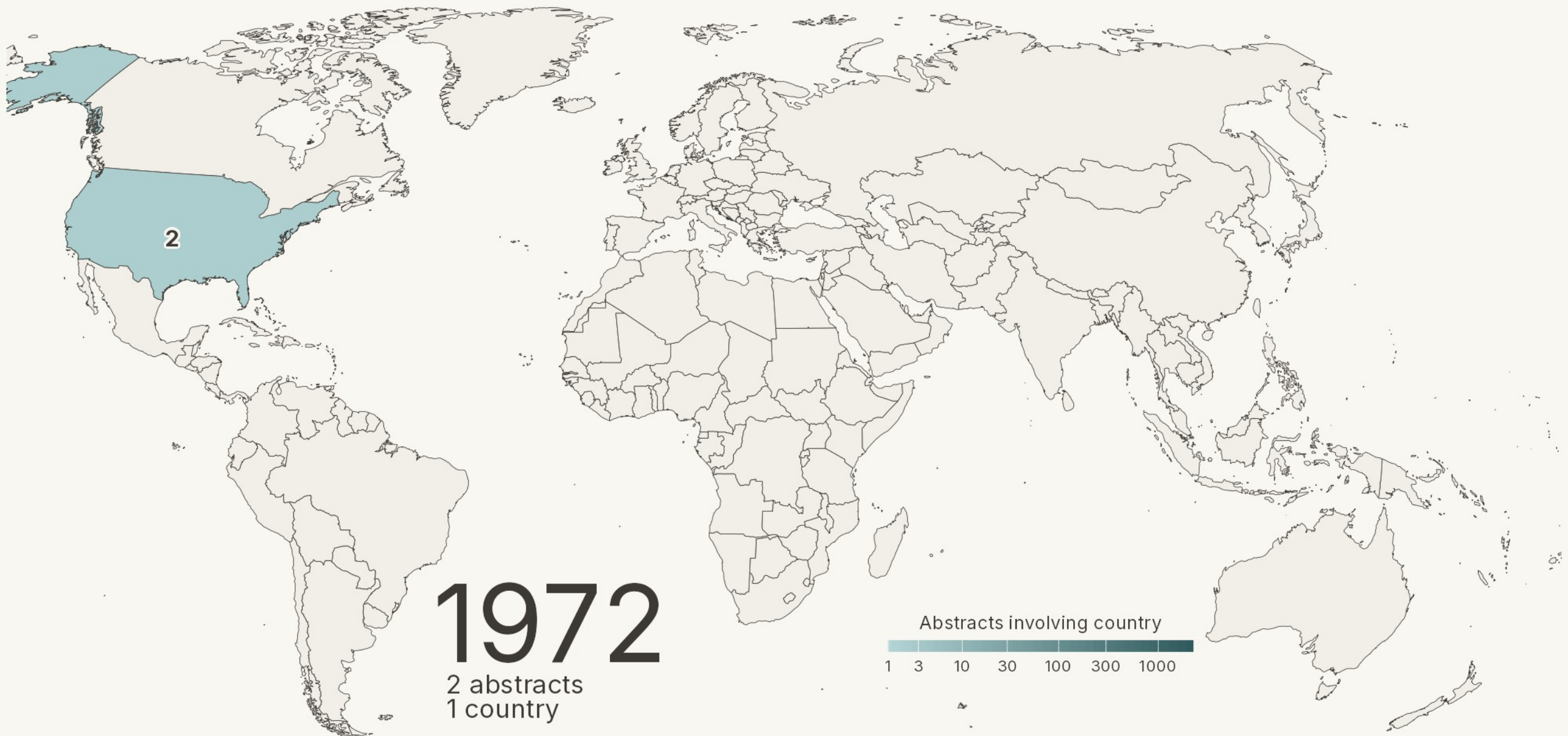
Recovering undetermined cases:

- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.

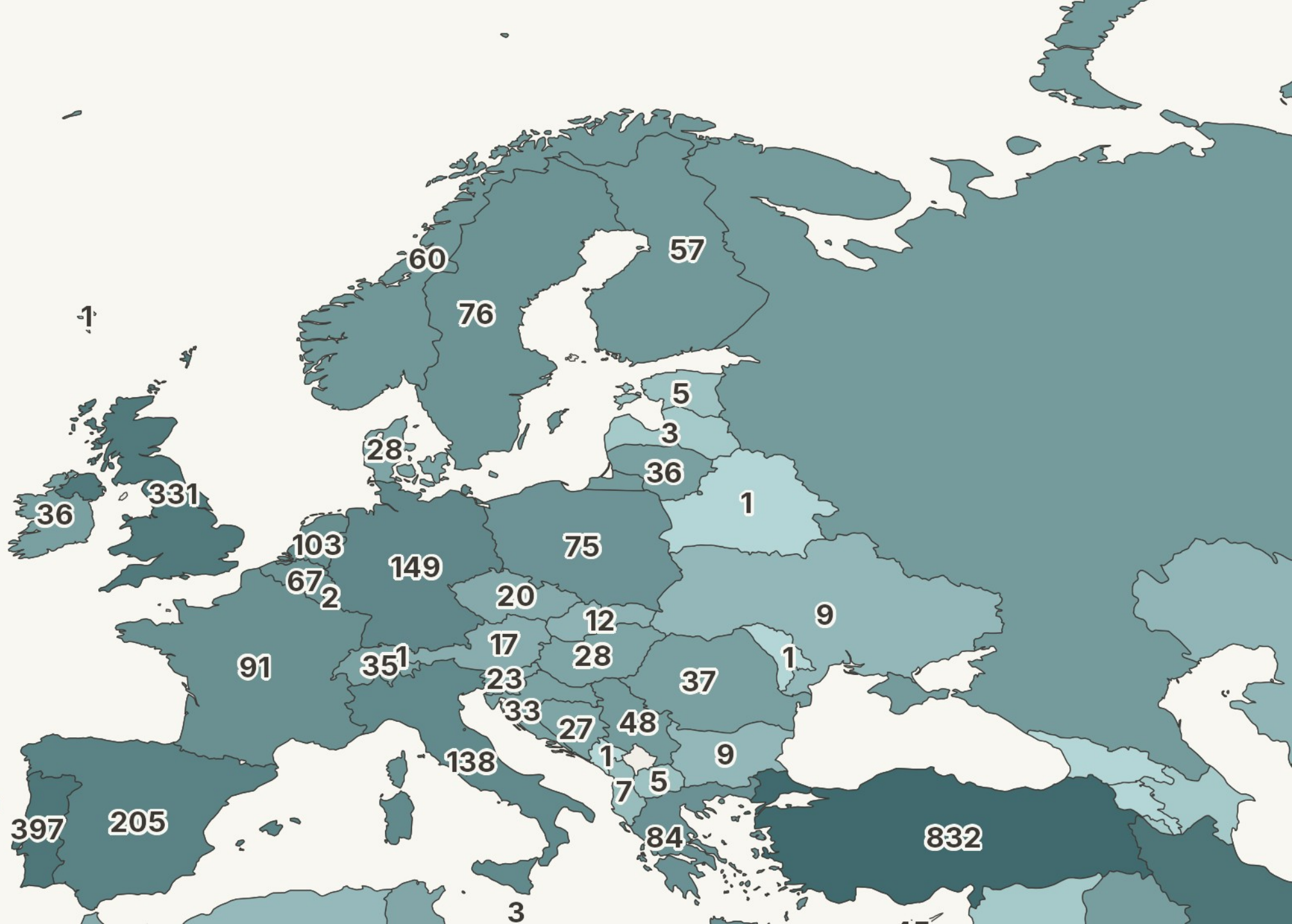
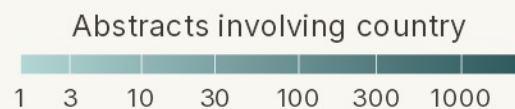


And mapped onto the globe — the geographic spread of the literature over five decades.



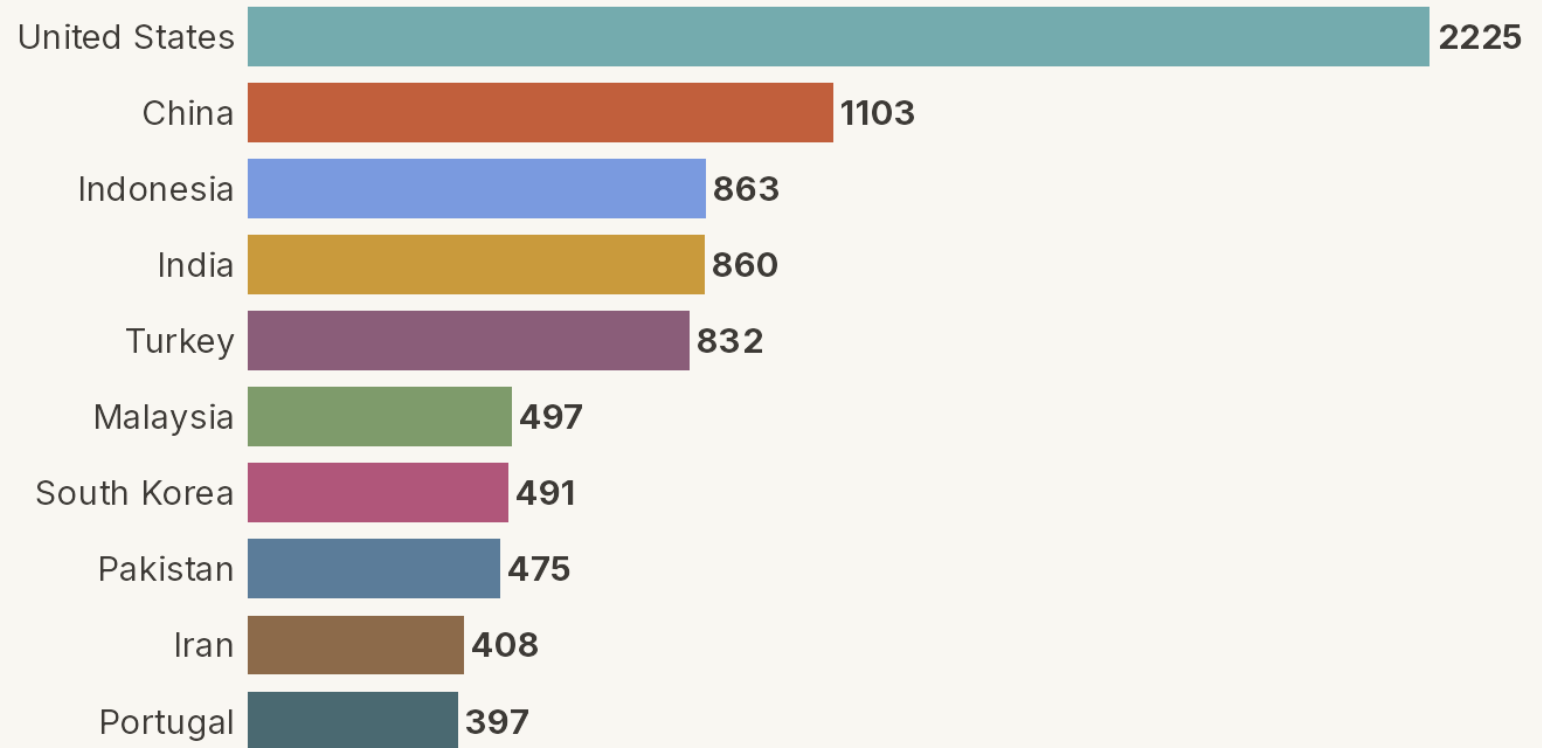


2025



Top 10 Countries by Cumulative Samples

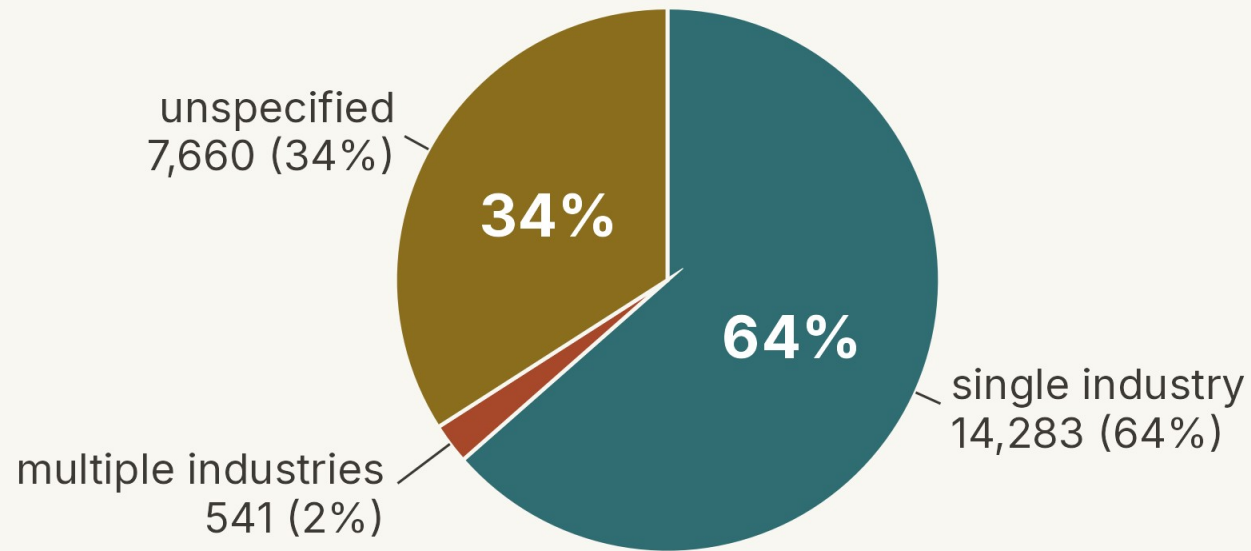
2025



What kinds of workplaces does this literature actually study? So far we have looked at growth and geography. Now we turn to the sampled settings themselves — the industries in which these commitment studies were conducted.

Industry — coverage in three tiers

Full coded sample, N = 22,484



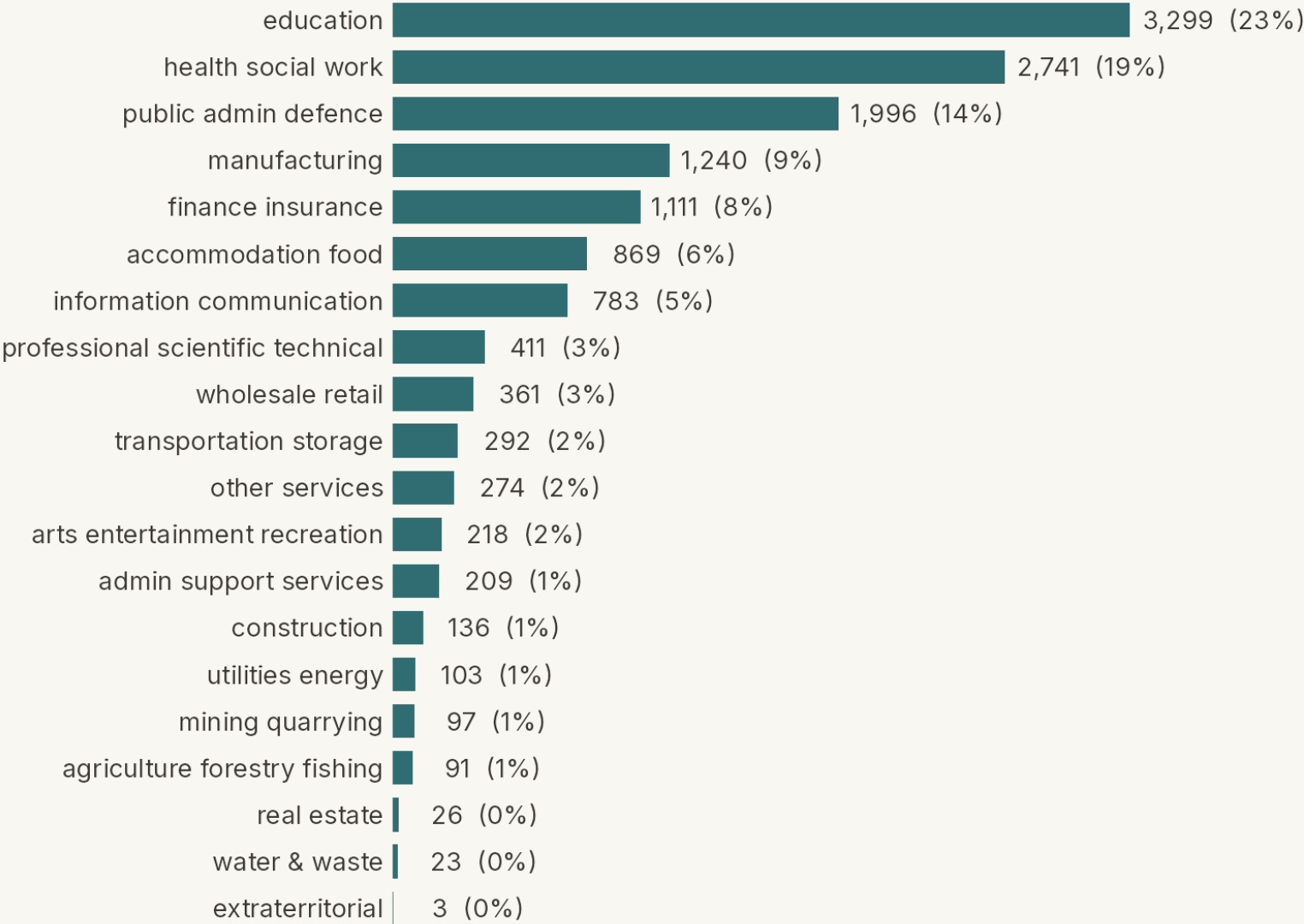
Industry taxonomy — ISIC Rev. 4

The UN's standard industry taxonomy — 21 letter-coded sections, A (agriculture, forestry & fishing) through U (extraterritorial organizations).

Each study was tagged by the industry of its sampled workplace, with two fallbacks: “multiple industries” for samples spanning several sections, and “unspecified” when the abstract named no setting at all.

Industry — distribution among placed studies

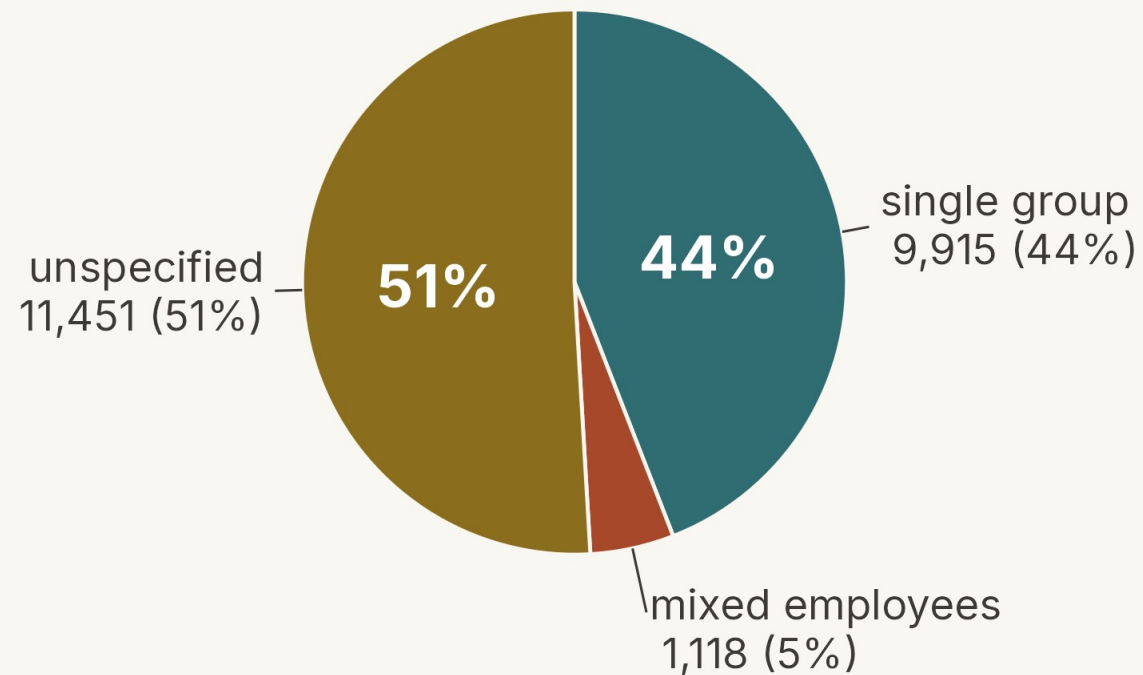
Single-industry studies, N = 14,283



Industries told us where these commitment studies were conducted. Occupations tell us who was sampled. The two questions are distinct: a hospital can sample nurses, managers, or every employee, and each choice carries a different inferential warrant.

Occupation — coverage in three tiers

Full coded sample, N = 22,484



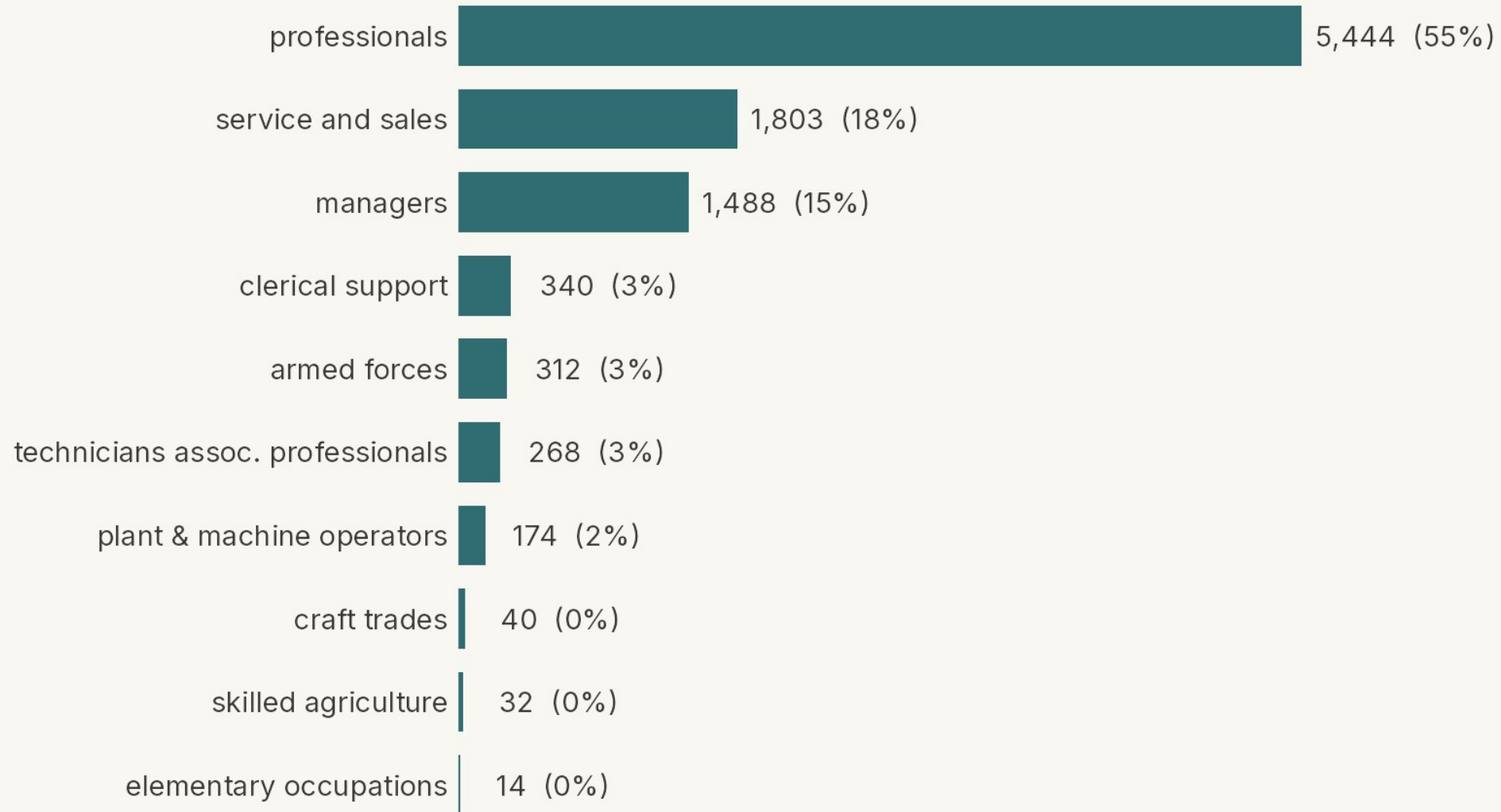
Occupation taxonomy — ISCO-08

The ILO's standard occupation taxonomy (the counterpart to ISIC) — 10 major groups, 0 (armed forces) through 9 (elementary occupations).

Each study was tagged by the occupation of its sample, with two fallbacks: “mixed employees” for samples spanning several major groups, and “unspecified” when no occupation was named.

Occupation — distribution among single-group studies

Single-group studies, N = 9,915

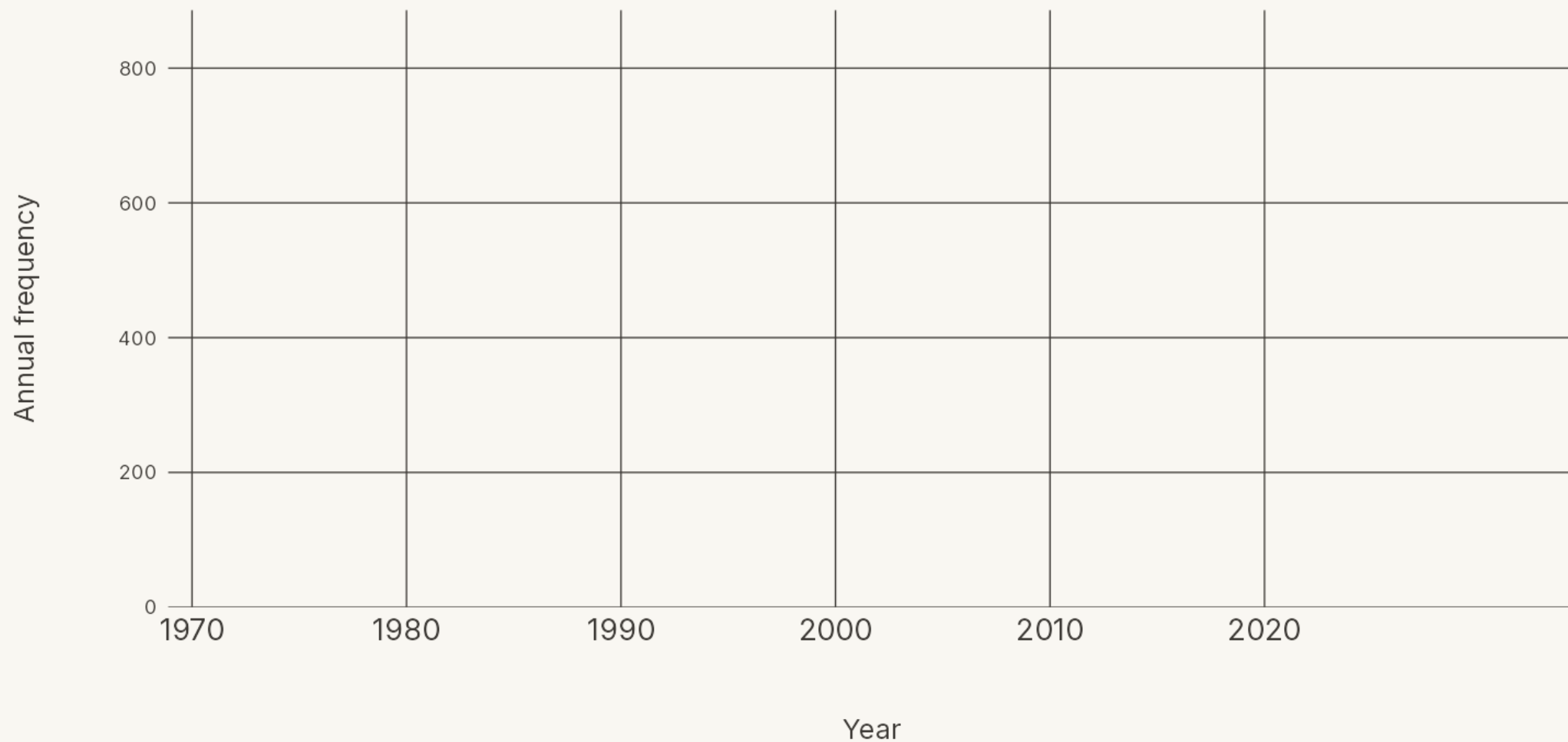




Finally, how is “organizational commitment” actually used in the literature? It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

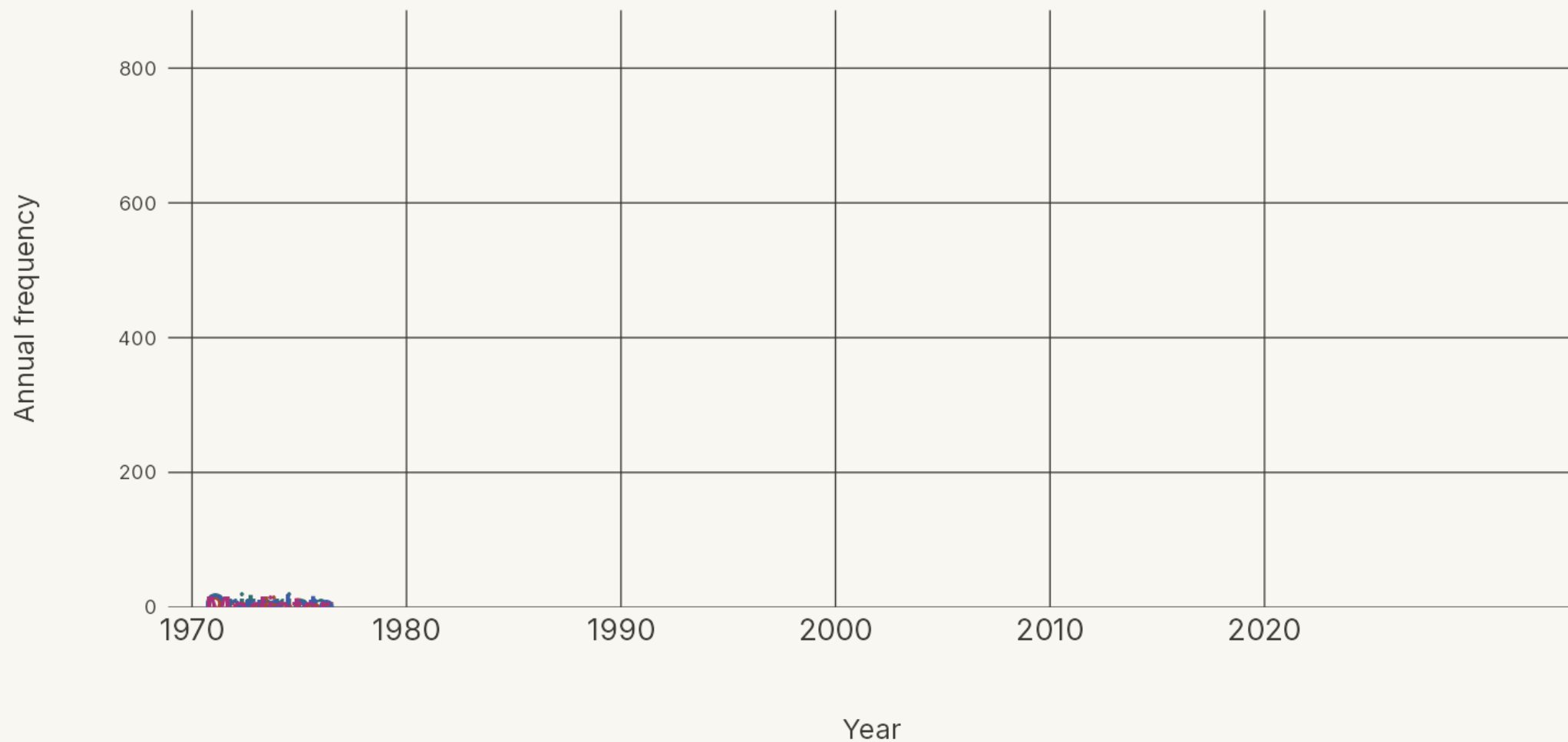
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



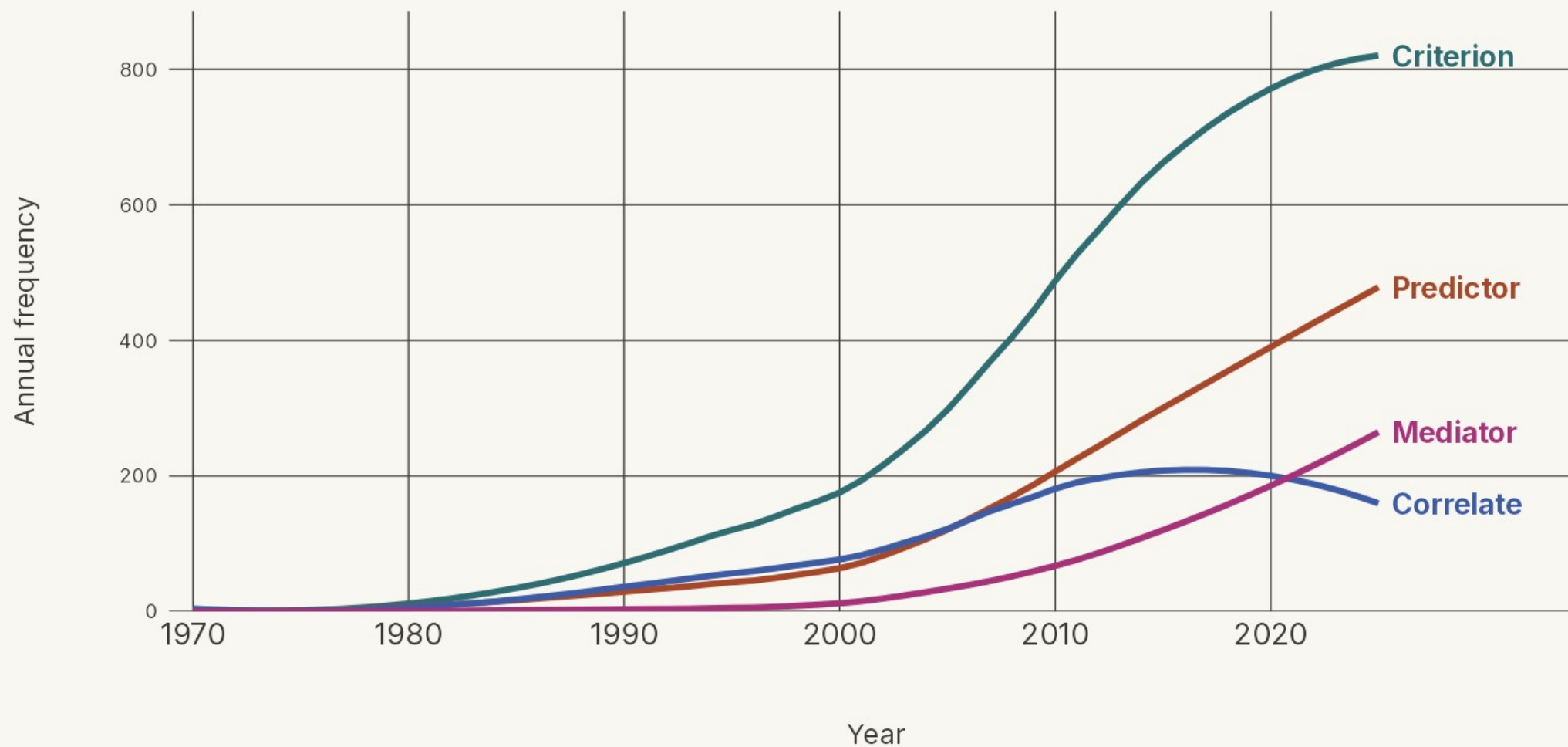
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



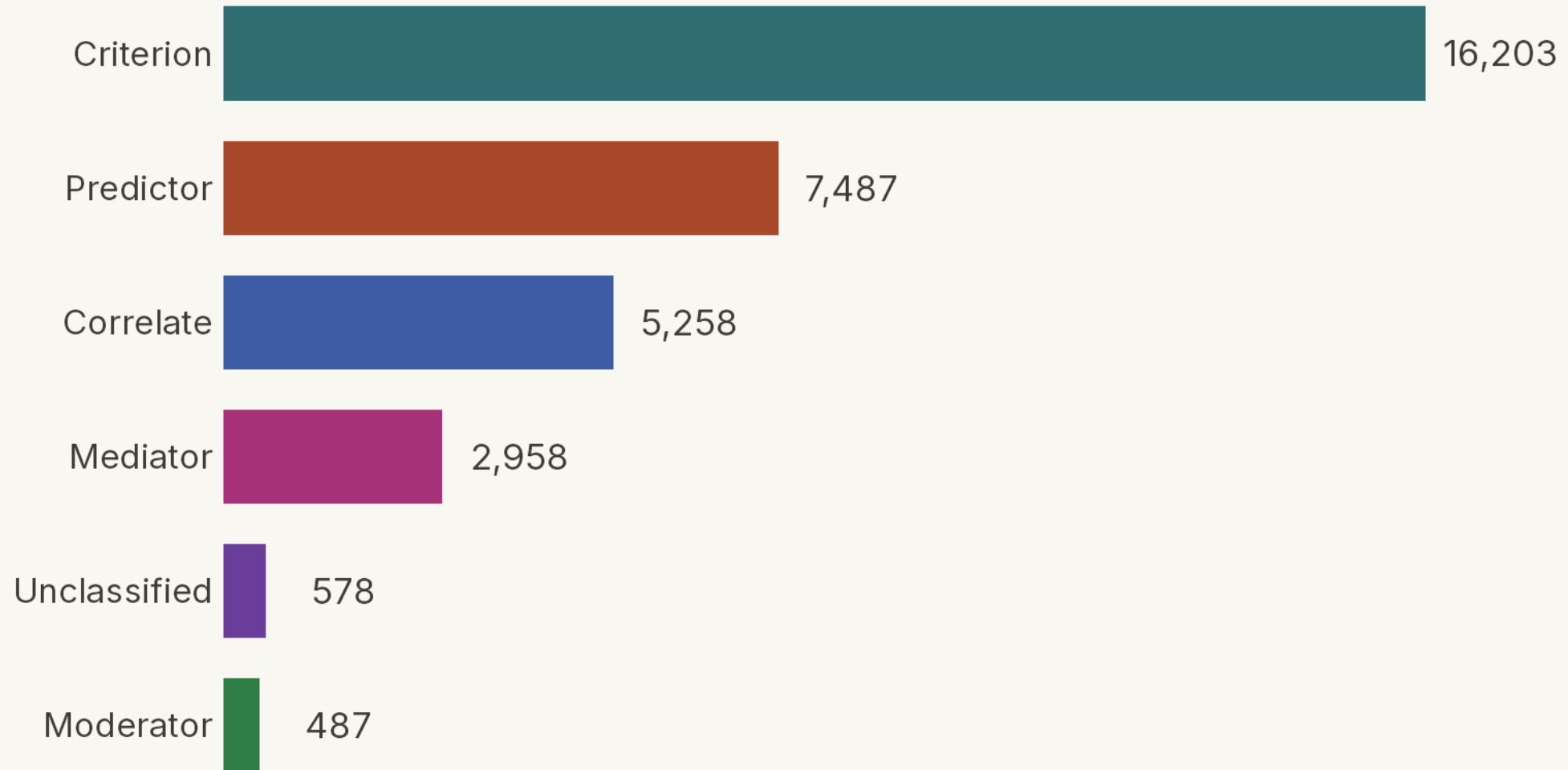
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



How 'organizational commitment' is used — all six roles

Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded



And which specific variables show up in those roles?

Across the full coded set we can look at three perspectives — what predicts commitment, what commitment predicts, and what is merely correlated with it when the study's framing doesn't assert a direction.

How did we normalize the variable labels?

Light normalization — merging only plural/spelling variants, before construct grouping

1

Extract the raw labels

Split the semicolon-separated snake_case tags coded for each study; trim whitespace and drop blanks — one label per variable mention.

2

Merge plural / spelling variants

De-pluralize (...ies → ...y, a trailing s dropped) — but only when the singular is itself an already-coded label, so no merge is ever invented. The commoner spelling becomes the label.

3

Stop before meaning

No semantic or construct grouping yet — that is the later coauthor pass. Every merge is written to an audit file, so the step stays fully checkable.

Conservative by design — every merge is verified against the coded labels and logged. A cross-language unification step comes next, then the distinct-variable counts.

One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

■ As written across languages

job satisfaction

工作满意度

satisfação no trabalho

iş tatmini

satisfacción laboral

satisfaction au travail

직무 만족

الرضا الوظيفي

English

Chinese

Portuguese

Turkish

Spanish

French

Korean

Arabic

■ The unified label

job_satisfaction

one English label · **2,085** studies pooled

The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

How many distinct variables did we extract?

Light spelling normalization only — before construct grouping

■ Predictors of commitment

14,394

distinct variables · light model

15,838

Studies named ≥1

36,697

Total mentions

■ Outcomes of commitment

3,324

distinct variables · light model

7,462

Studies named ≥1

10,290

Total mentions

Light normalization is cosmetic — it merges only spellings and plurals; the real reduction comes from construct grouping.

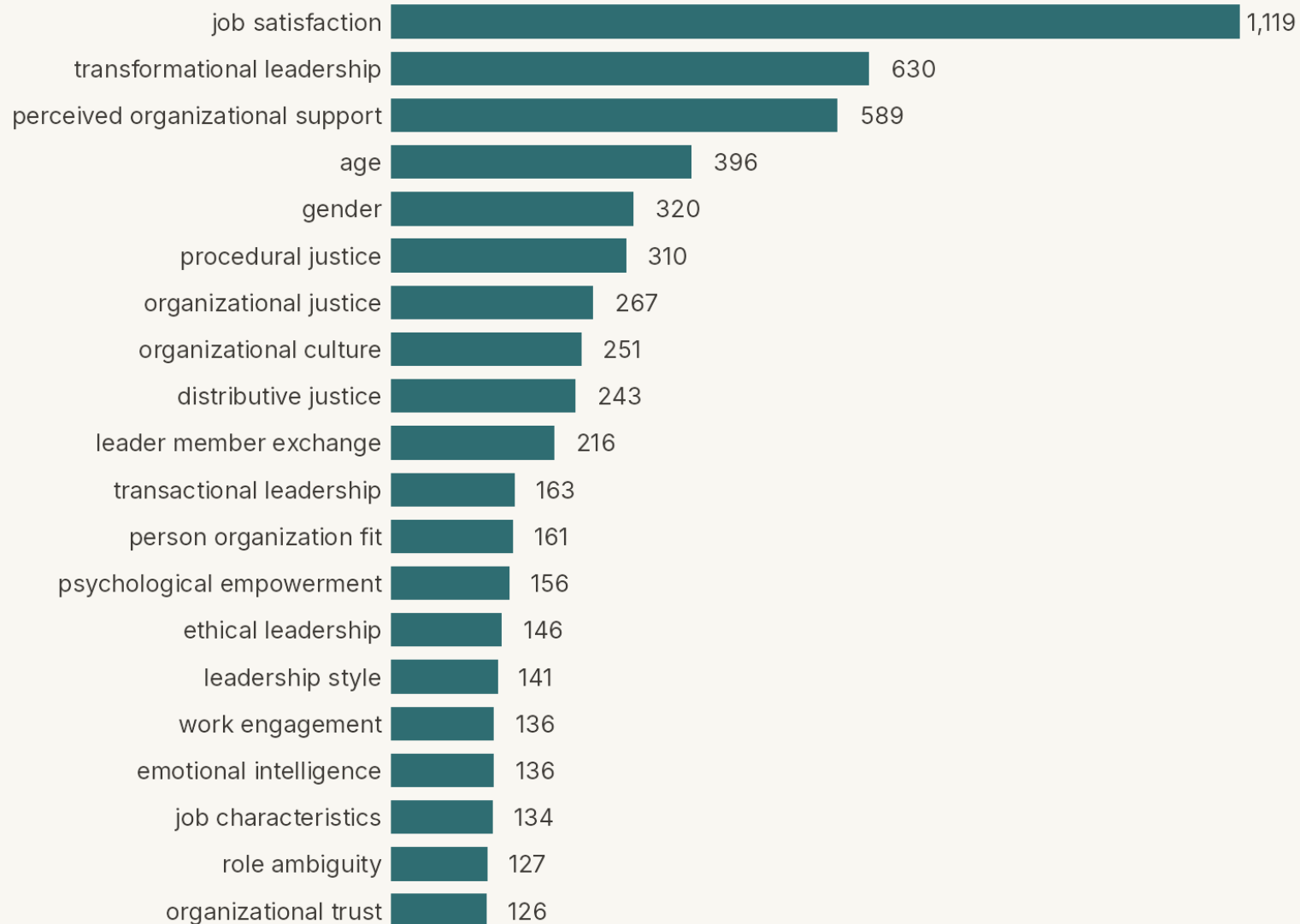
Top 20 correlates of organizational commitment

Across 4,365 studies naming ≥ 1 correlate (of 22,484 coded) — bars count distinct studies



Top 20 predictors of organizational commitment

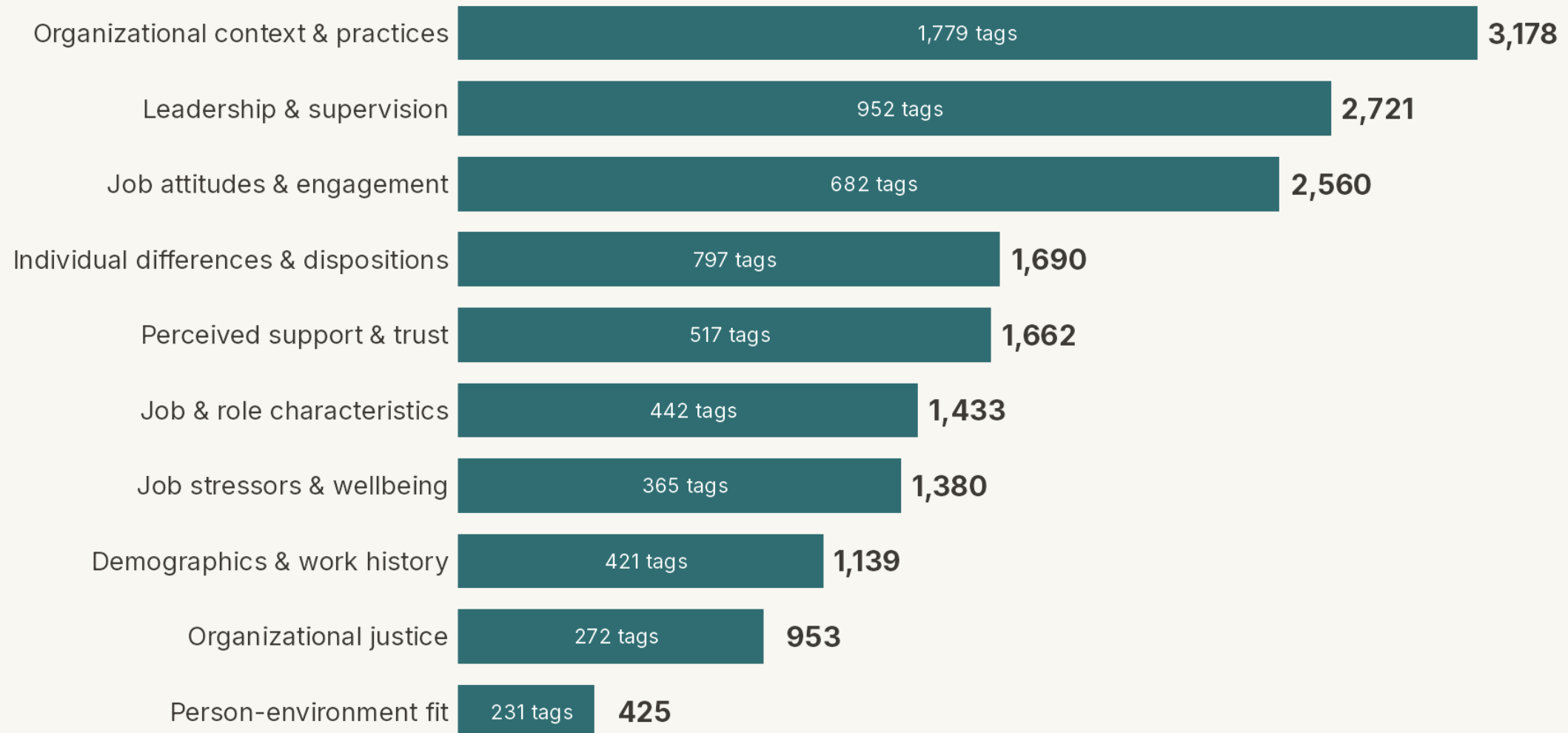
Across 15,838 studies naming ≥ 1 predictor (of 22,484 coded) — bars count distinct studies



Predictors of organizational commitment, in ten categories

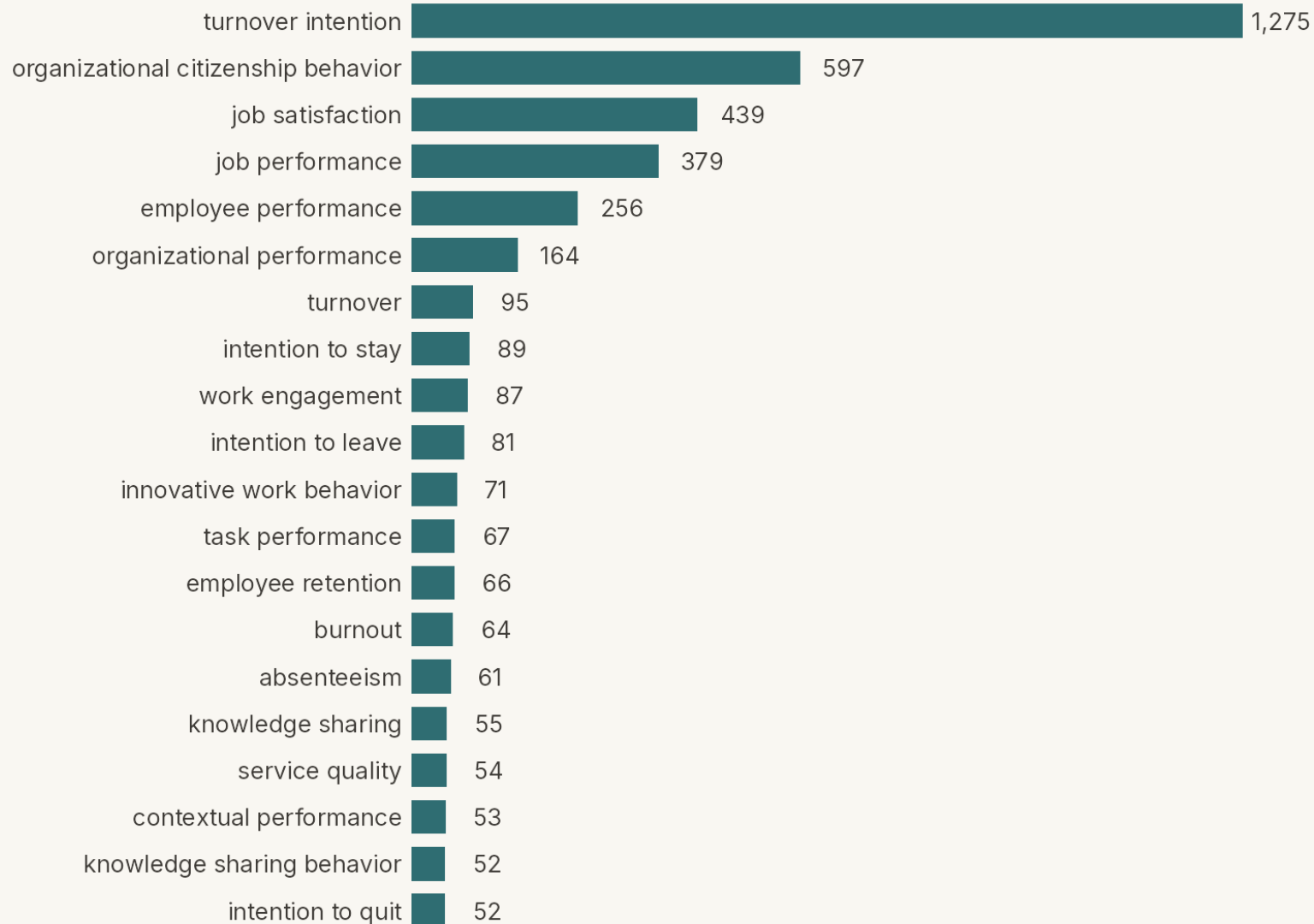
Bar & end label = distinct studies naming ≥ 1 predictor in the category.

Inside each bar = number of variables (tags) collapsed into it. ("Other" not shown.)



Top 20 outcomes of organizational commitment

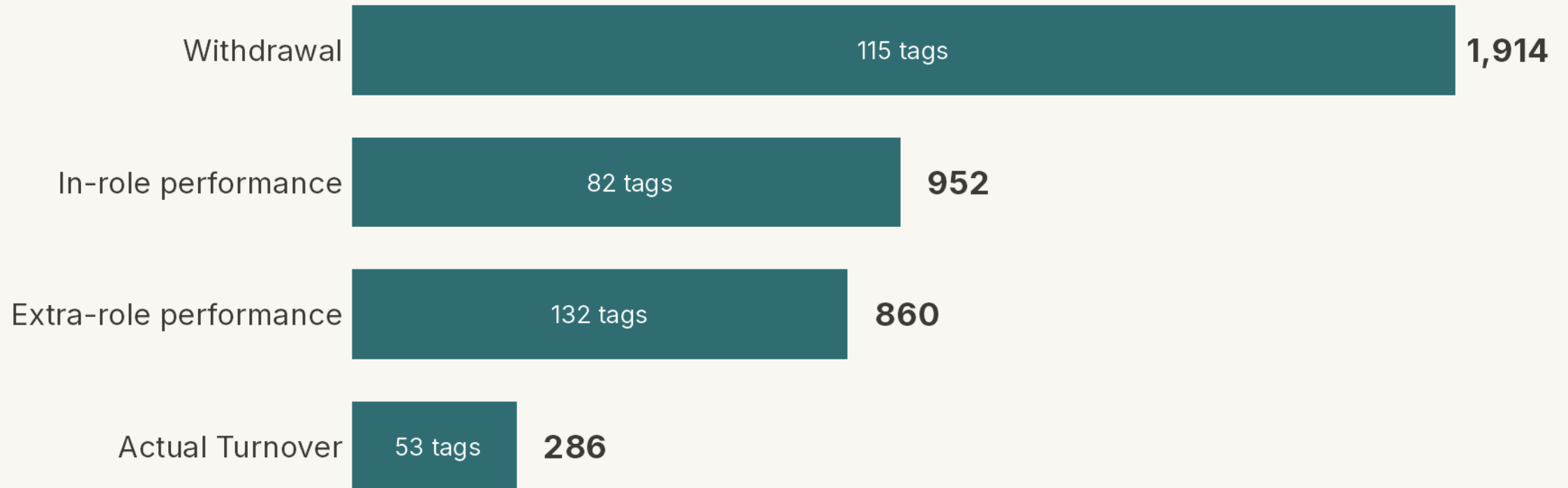
Across 7,462 studies naming ≥ 1 outcome (of 22,484 coded) — bars count distinct studies



Outcomes of organizational commitment, in four families

Bar & end label = distinct studies naming ≥ 1 outcome in the family.

Inside each bar = number of variables (tags) collapsed into it.



FROM ABSTRACTS TO ANALYSIS

Next Steps

The work ahead, in four steps.

How accurate is AI coding? — from prior research

The labour-intensive part is the coding itself — reading each study and pulling out every sample size, mean, correlation, and reliability by hand. In earlier published validation, the AI was compared against the expert human teams who did exactly that.

■ Data extraction — the coding itself

97.0%

AI accuracy — vs **92.1%** for expert human coders

4 meta-analyses · 632 studies · 20,476 data points

■ What the AI is being compared to

One of the four meta-analyses, single-coded by hand:

130 hrs → **4 hrs**
one coder ≈ 32× faster the AI

12,025 numbers from 390 studies — one coder, about 20 min a study.

- **Screening** — deciding which studies belong — was just as strong: the AI caught **97.2%** of the relevant studies (humans 91.6%) across 30,150 decisions.

Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1

Download the articles

Retrieve the full texts behind the coded abstracts.

2

Code the articles

Extract effect sizes and study details from each.

3

Formalize the variable taxonomies

Finalize the construct groupings from these results.

4

Begin analyses

Run the meta-analytic models.

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Foundational works

1 OF 2

The cited-reference search retrieved every record citing any of these 14 works

Allen, N. J., & Meyer, J. P. (1990). The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology*, 63(1), 1–18.

Allen, N. J., & Meyer, J. P. (1996). Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior*, 49(3), 252–276.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology*, 53(1), 39–52.

Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014). The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology*, 99(2), 222–238.

Meyer, J. P., & Allen, N. J. (1984). Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology*, 69(3), 372–378.

Meyer, J. P., & Allen, N. J. (1991). A three-component conceptualization of organizational commitment. *Human Resource Management Review*, 1(1), 61–89.

Meyer, J. P., & Allen, N. J. (1997). *Commitment in the workplace: Theory, research, and application*. Sage Publications.

Foundational works

2 OF 2

The cited-reference search retrieved every record citing any of these 14 works

Meyer, J. P., Allen, N. J., & Smith, C. A. (1993). Commitment to organizations and occupations: Extension and test of a three-component conceptualization. *Journal of Applied Psychology*, 78(4), 538–551.

Meyer, J. P., Stanley, D. J., Herscovitch, L., & Topolnytsky, L. (2002). Affective, continuance, and normative commitment to the organization: A meta-analysis of antecedents, correlates, and consequences. *Journal of Vocational Behavior*, 61(1), 20–52.

Mowday, R. T., Porter, L. W., & Steers, R. M. (1982). *Employee–organization linkages: The psychology of commitment, absenteeism, and turnover*. Academic Press.

Mowday, R. T., Steers, R. M., & Porter, L. W. (1979). The measurement of organizational commitment. *Journal of Vocational Behavior*, 14(2), 224–247.

O'Reilly, C. A., III, & Chatman, J. (1986). Organizational commitment and psychological attachment: The effects of compliance, identification, and internalization on prosocial behavior. *Journal of Applied Psychology*, 71(3), 492–499.

Porter, L. W., Steers, R. M., Mowday, R. T., & Boulian, P. V. (1974). Organizational commitment, job satisfaction, and turnover among psychiatric technicians. *Journal of Applied Psychology*, 59(5), 603–609.

Powell, D. M., & Meyer, J. P. (2004). Side-bet theory and the three-component model of organizational commitment. *Journal of Vocational Behavior*, 65(1), 157–177.